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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 6

Title: Creating HR analytics dashboards

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Creating HR Analytics Dashboard using Power BI

Problem Statement

A multinational organization requires an HR Analytics Dashboard to gain valuable insights into employee demographics, workforce distribution, employee attrition, salary trends, and employee performance. The objective is to develop an interactive dashboard using the IBM HR Analytics Employee Attrition Dataset that enables HR managers to monitor workforce statistics, identify attrition patterns, analyze employee characteristics, and support strategic workforce planning and decision-making.

Dataset Used: IBM HR Analytics Employee Attrition Dataset (Kaggle)

Methodology / Steps Followed

1. Downloaded the **IBM HR Analytics Employee Attrition Dataset** from Kaggle.
2. Imported the dataset into **Power BI Desktop** using the **Get Data** option.
3. Verified data types and checked for duplicate records and missing values in **Power Query Editor**.
4. Loaded the cleaned dataset into Power BI for analysis.
5. Created **DAX measures** to calculate Total Employees, Active Employees, Attrition Count, Attrition Rate, Average Age, and Average Monthly Income.
6. Created **KPI Cards** to display important HR metrics.
7. Developed interactive visualizations such as **Bar Charts, Donut Charts, Line Charts, Treemap, Matrix, Gauge Chart, and Column Charts**.
8. Analyzed employee demographics, department-wise distribution, attrition, salary trends, education field, and work-life balance.
9. Added **Slicers** for Department, Job Role, OverTime, and Total Working Years to enable interactive filtering.
10. Formatted the dashboard with appropriate titles, colors, and layouts to create a professional and user-friendly HR Analytics Dashboard.

Output

- Importing IBM HR Analytics Dataset

Get Data (Power Query)

Preview file data

File path: C:\Users\AMD\Downloads\archive (30)\WA_Fn-UseC_-HR-Employee-Attrition.csv

File origin: 65001: Unicode (UTF-8) Delimiter: Comma Data type detection: Based on first 200 rows

123 Age	Attrition	BusinessTravel	123 DailyRate	Department	123 DistanceFromHome	123 Education	EducationField	123 EmployeeCount	123 EmployeeNumber	123 EnvironmentSatisfaction	Gender	123 H
41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	2	Female	
49	No	Travel_Frequently	279	Research & Developm...	8	1	Life Sciences	1	2	3	Male	
37	Yes	Travel_Rarely	1373	Research & Developm...	2	2	Other	1	4	4	Male	
33	No	Travel_Frequently	1392	Research & Developm...	3	4	Life Sciences	1	5	4	Female	
27	No	Travel_Rarely	591	Research & Developm...	2	1	Medical	1	7	1	Male	
32	No	Travel_Frequently	1005	Research & Developm...	2	2	Life Sciences	1	8	4	Male	
59	No	Travel_Rarely	1324	Research & Developm...	3	3	Medical	1	10	3	Female	
30	No	Travel_Rarely	1358	Research & Developm...	24	1	Life Sciences	1	11	4	Male	
38	No	Travel_Frequently	216	Research & Developm...	23	3	Life Sciences	1	12	4	Male	
36	No	Travel_Rarely	1299	Research & Developm...	27	3	Medical	1	13	3	Male	
35	No	Travel_Rarely	809	Research & Developm...	16	3	Medical	1	14	1	Male	
29	No	Travel_Rarely	153	Research & Developm...	15	2	Life Sciences	1	15	4	Female	
31	No	Travel_Rarely	670	Research & Developm...	26	1	Life Sciences	1	16	1	Male	
34	No	Travel_Rarely	1346	Research & Developm...	19	2	Medical	1	18	2	Male	
28	Yes	Travel_Rarely	103	Research & Developm...	24	3	Life Sciences	1	19	3	Male	
29	No	Travel_Rarely	1389	Research & Developm...	21	4	Life Sciences	1	20	2	Female	
32	No	Travel_Rarely	334	Research & Developm...	5	2	Life Sciences	1	21	1	Male	
22	No	Non-Travel	1123	Research & Developm...	16	2	Medical	1	22	4	Male	
53	No	Travel_Rarely	1219	Sales	2	4	Life Sciences	1	23	1	Female	
28	No	Travel_Rarely	271	Research & Developm...	2	3	Life Sciences	1	24	4	Male	

Back Cancel Add table using examples Load

► Data Validation and Cleaning

Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Any Use First Row as Header Replace Values Transform

Remove Top Rows Remove Bottom Rows Remove Alternate Rows Remove Duplicates Remove Blank Rows Remove Errors

123 DailyRate	ent	123 Dist
1373	Research & Development	
1392	Research & Development	
591	Research & Development	
1005	Research & Development	
1324	Research & Development	
1358	Research & Development	

= Table.TransformColumnTypes(#"Changed column ty			
	123 Age	ABC Attrition	ABC BusinessTravel
1	1.2	Decimal Number	Travel_Rarely
2	\$	Fixed decimal number	Travel_Frequently
3	123	Whole Number	Travel_Rarely
4	%	Percentage	Travel_Frequently
5		Date/Time	Travel_Rarely
6		Date	Travel_Frequently
7		Time	Travel_Rarely
8		Date/Time/Timezone	Travel_Rarely
9		Duration	Travel_Frequently
10	ABC	Text	Travel_Rarely
11		True/False	Travel_Rarely
12		Binary	Travel_Rarely
13		Using Locale...	Travel_Rarely
14			Travel_Rarely
15	28	Yes	Travel_Rarely

► Creating DAX Measures

```
1 Active Employees =
2 [Total Employees]-[Attrition Count]
```

```
1 Attrition Count =
2 CALCULATE(
3 COUNT(HR_Data[EmployeeNumber]),
4 HR_Data[Attrition]="Yes"
5 )
```

```
1 Attrition Rate =
2 DIVIDE(
3 [Attrition Count],
4 [Total Employees],
5 0
6 )
```

```
1 Average Job Satisfaction =
2 AVERAGE(HR_Data[JobSatisfaction])
```

```
1 Average Monthly Income =
2 AVERAGE(HR_Data[MonthlyIncome])
```

```
1 Female Employees =
2 CALCULATE(
3 COUNT(HR_Data[EmployeeNumber]),
4 HR_Data[Gender]="Female"
5 )
```

```

1 Overtime Attrition =
2 CALCULATE(
3 COUNT(HR_Data[EmployeeNumber]),
4 HR_Data[OverTime]="Yes",
5 HR_Data[Attrition]="Yes"
6 )

```

```

1 Total Employees =
2 COUNT(HR_Data[EmployeeNumber])

```

```

1 Overtime Employees =
2 CALCULATE(
3 COUNT(HR_Data[EmployeeNumber]),
4 HR_Data[OverTime]="Yes"
5 )

```

► Building Interactive HR Analytics Dashboard



► Employee Demographics Dashboard



Results

- ☐ The IBM HR Analytics dataset was successfully imported, cleaned, and analyzed using Power BI.
- ☐ KPI cards provided a summary of employee count, attrition, age, and salary.
- ☐ Charts highlighted workforce distribution by department, gender, education field, and job role.
- ☐ Attrition analysis identified employee turnover across age groups and overtime status.
- ☐ Salary trends and experience analysis helped compare income across job roles and years at the company.
- ☐ Interactive slicers enabled users to explore HR insights dynamically and support better workforce decision-making.

Business Insights

- The organization consists of 1,470 employees, providing a comprehensive overview of the workforce.
- The overall employee attrition rate is approximately 16%, indicating that employee retention should remain a strategic focus.
- Research & Development employs the highest number of employees, making it the largest department within the organization.

- Employees aged 26–35 years form the largest workforce segment and also contribute the highest attrition count.
- Employees working overtime exhibit higher attrition compared to those who do not work overtime.
- Managers receive the highest average monthly income among all job roles, while entry-level positions receive comparatively lower salaries.
- The majority of employees belong to the Life Sciences and Medical education fields, reflecting the organization's talent composition.
- Married employees represent the largest proportion of the workforce, followed by Single employees.
- Workforce experience is concentrated among employees with fewer years at the company, indicating a relatively young employee base.
- Department-wise gender distribution demonstrates a balanced workforce while highlighting opportunities to monitor diversity and inclusion initiatives.
- Interactive dashboard filters enable HR managers to perform detailed workforce analysis and support evidence-based decision-making.

Conclusion

This experiment demonstrated the development of an interactive HR Analytics Dashboard using Microsoft Power BI and the IBM HR Analytics Employee Attrition Dataset. The dashboard integrates multiple KPI cards, DAX measures, charts, and interactive slicers to analyze employee demographics, department-wise workforce distribution, salary trends, employee attrition, education background, and work-life characteristics. The dashboard provides meaningful business insights that support workforce planning, employee retention strategies, and informed HR decision-making.

Learning Outcomes

- ✓ Understand the role of HR Analytics in workforce planning and decision-making.
- ✓ Import and prepare the IBM HR Analytics Dataset in Power BI.
- ✓ Verify data quality by checking data types, duplicate records, and missing values.
- ✓ Create DAX Measures for HR KPIs.

- ✓ Design KPI Cards to summarize workforce statistics.
- ✓ Build Bar Charts, Donut Charts, Treemaps, Line Charts, Gauge Charts, Matrix visuals, and Column Charts.
- ✓ Analyze employee attrition across departments, age groups, and overtime status.
- ✓ Compare salary distribution among different job roles.
- ✓ Analyze employee demographics including gender, education field, and marital status.
- ✓ Implement slicers for interactive dashboard filtering.
- ✓ Apply formatting techniques to design a professional HR Analytics Dashboard.
- ✓ Gain practical experience in developing business intelligence dashboards for HR management.

GitHub Link: https://github.com/lavanyae4319/Data_Analytics_Assignments